

Network Security Monitoring Made Easy with Zeek



Zeek is an open source network security monitoring tool. This article discusses its capabilities, and gives a short tutorial on how to get started with it.

A network is a collection of computers, servers, wired or wireless devices or any other devices connected to each other to allow data sharing. For example, the Internet is a network where billions and millions of devices are connected all over the globe. A smaller example is our home network, which connects our laptop, mobile, hubs and router.

Network analysis is a method used to examine the data that is transferred between two devices in a network. It helps to understand the architecture of the network and also investigates any malicious activity. There are several open source tools used to analyse network traffic. One among them is Zeek.

Zeek: An open source network monitoring tool

Zeek is a passive, open source network traffic analyser tool. Many companies and operators use Zeek as a NSM (network security monitor), as it is very

helpful for investigating suspicious or malicious activity in the network. This tool also supports a wide range of traffic analysis tasks beyond the cybersecurity domain, including performance measurement and troubleshooting.

Zeek logs network activity in a separate file. These logs include all the data like HTTP sessions, requested URIs, MIME types, server responses, DNS requests, SSL certificates, key content of SMTP sessions, and much more. All the data is written in a well-structured, tab separated or JSON format.

In addition to this, Zeek can perform a few special tasks like

extracting files from HTTP sessions, detecting malware, detecting SSH brute forcing, and validating SSL certificates.

Installing Zeek

To install Zeek on a Debian based Linux system, follow the steps given below and execute the commands as shown.

Step 1: Add the Zeek repository to the *sources.list* file by executing the following command:

```
echo 'deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_20.04/ ' | sudo tee /etc/apt/sources.list.d/security:zeek.list
```

```
tridev@Tridev-Reddy:~$
$ echo 'deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_20.04/ ' |
sudo tee /etc/apt/sources.list.d/security:zeek.list
deb http://download.opensuse.org/repositories/security:/zeek/xUbuntu_20.04/
$ curl -fsSL https://download.opensuse.org/repositories/security:zeek/xUbuntu_20.04/Release.key | gpg --dearmor | sudo tee /etc/apt/trusted.gpg.d/security_zeek.gpg > /dev/null
~ Tridev-Reddy
```

Figure 1: Adding Zeek repo to *sources.list*